

# Sensory Robotic Arm

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## Sensory Robotic Arm

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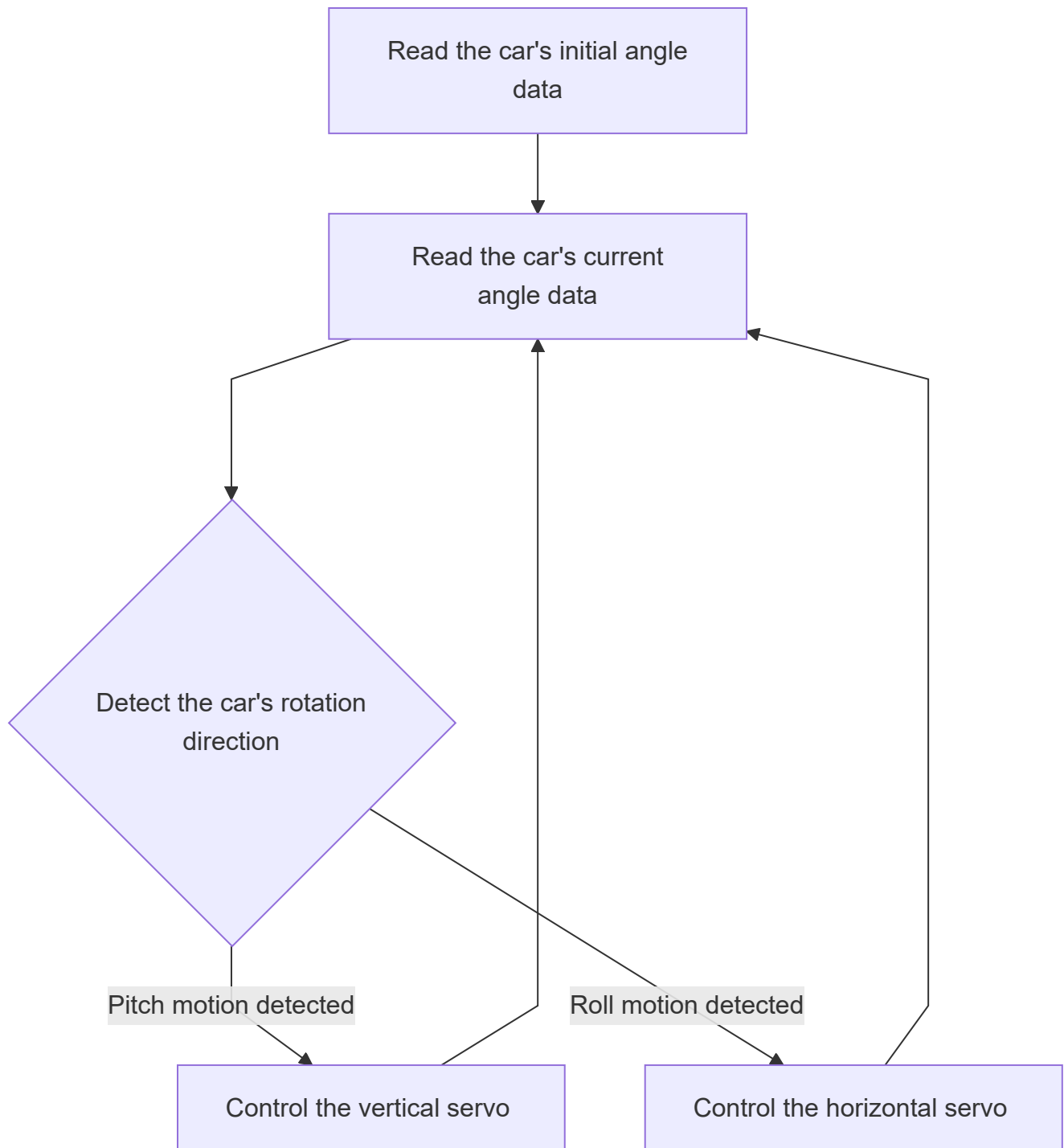
## 1. Experiment Purpose

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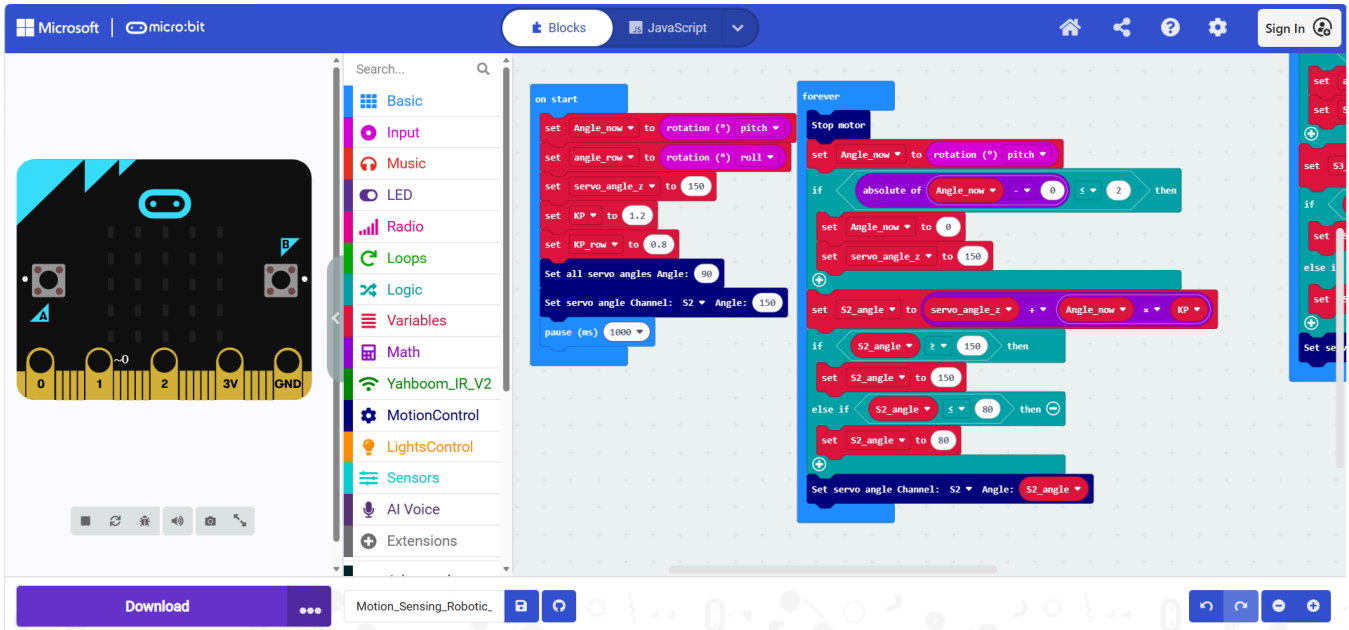
In this section, we will learn to use the accelerometer sensor to realize motion-sensing control of the robotic arm. By tilting the Micro:bit, you can control the pitch and roll angles of the robotic arm, achieving motion-sensing manipulation.

## 2. Implementation Principle

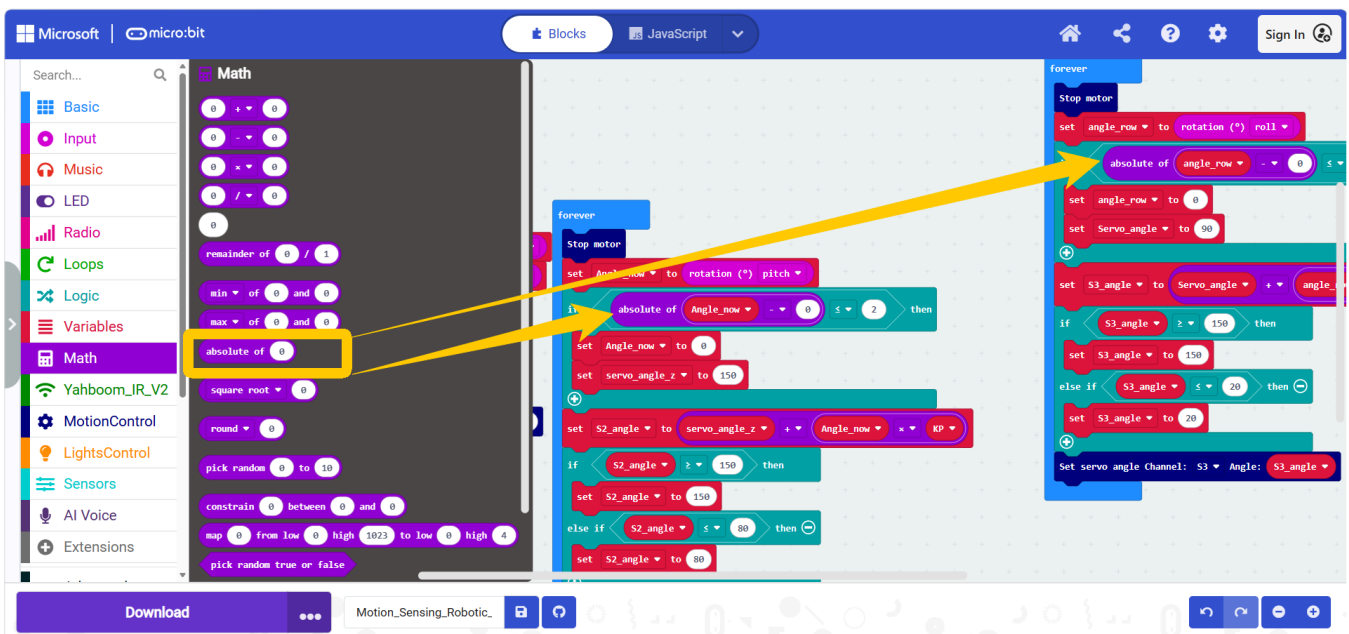
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The program is as follows.



Taking the absolute value is located in the **Math** category.



The positions of other blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

## 4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `10.Motion_Sensing_Robotic_Arm.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

## 5. Experiment Phenomenon

It is recommended to move the car slowly.

At this time, lift up the front of the car, and the car will control the vertical servo to move downward.



Lift up the rear of the car, and the car will control the vertical servo to move upward.





Tilt the car body to the right, and the car will control the horizontal servo to move left.



Tilt the car body to the left, and the car will control the horizontal servo to move right.



